# Multinomial Distribution MLE: Part 1

← Back

Graded Assignment • 10 min

**Due** May 14, 11:59 PM PDT

## Overview

This assignment begins with some conceptual questions about Peter's derivation for the MLE of the Multinomial/Categorical distribution in the video you just watched. After this, you can compute the MLE for a data set in the Python lab and then check your work in Parts 2 and 3 of this assignment, respectively.

1. **In the derivation, Peter takes:**

1 point

$\frac{\partial \mathcal{L}}{\partial \mu_k} = 0$, **where** $\mathcal{L}(\vec{\mu}, \lambda) = \sum_{k=1}^{K} m_k log(\mu_k) + \lambda(\sum_{k=1}^{K} \mu_k - 1)$, **where** $log$ **is the natural logarithm** $ln$**, and evaluates this to be** $\frac{\partial \mathcal{L}}{\partial \mu_k} = 0 = \frac{m_k}{\mu_k} + \lambda$.

**In your own words, why do the sums drop out of the equation when we take the partial derivative with respect to** $\mu_k$**?**

**It's okay to say things in words- the goal is not for you to spend a lot of time figuring out how to format the text/equations here. Try to explain this in plain, simple, and concise language.**

**This understanding is vital, as we will leverage this property of partial derivatives extensively in this course.**

When you take the partial derivative with respect to a single specific variable, all other variables in the summation are treated as constants.Here is why each sum collapses:The first sum: Every term in this sum that does not contain the specific target variable has a derivative of 0. The only surviving term differentiates to m_k / u_k.The second sum: When distributing lambda, the only term containing the target variable is lambda * u_k. Since the derivative of lambda * u_k with respect to u_k is simply lambda, all other constant terms disappear.

2. **At a high level, describe in your own words the steps we took to evaluate the**

1 point